Calculus II **Worksheet** Name:

1. A cylindrical tank has a radius of 5 ft and a height of 13 ft. Set up an integral to determine how much work is done in emptying it if
   (a) It is filled with molasses (which weighs 235 lb/ft^3).

   (b) It is filled to a depth of 9 ft and the molasses must be pumped out a 2 ft pipe at the top of the tank.

2. A conical tank has a radius of 5 ft and a height of 13 ft. Set up an integral to determine how much work is done in emptying it if
   (a) It is filled with milk (which weighs 64.5 lb/ft^3).

   (b) It is filled to a depth of 9 ft and the milk must be pumped out a 2 ft pipe at the top of the tank.

3. Discuss the following completely.
   (a) What is the fundamental idea of Chapter 6?

   (b) Suppose an object moves along the x-axis from $x = 2$ to $x = 5$ and is acted on by a force $F(x) = x^2$. Set up an integral to determine the amount of work done on the particle.

   (c) Use the fundamental idea of Chapter 6 to explain in complete detail why the work on the particle is given by your integral.